

Double Pendulum Using Euler-Lagrange Equations and RK4

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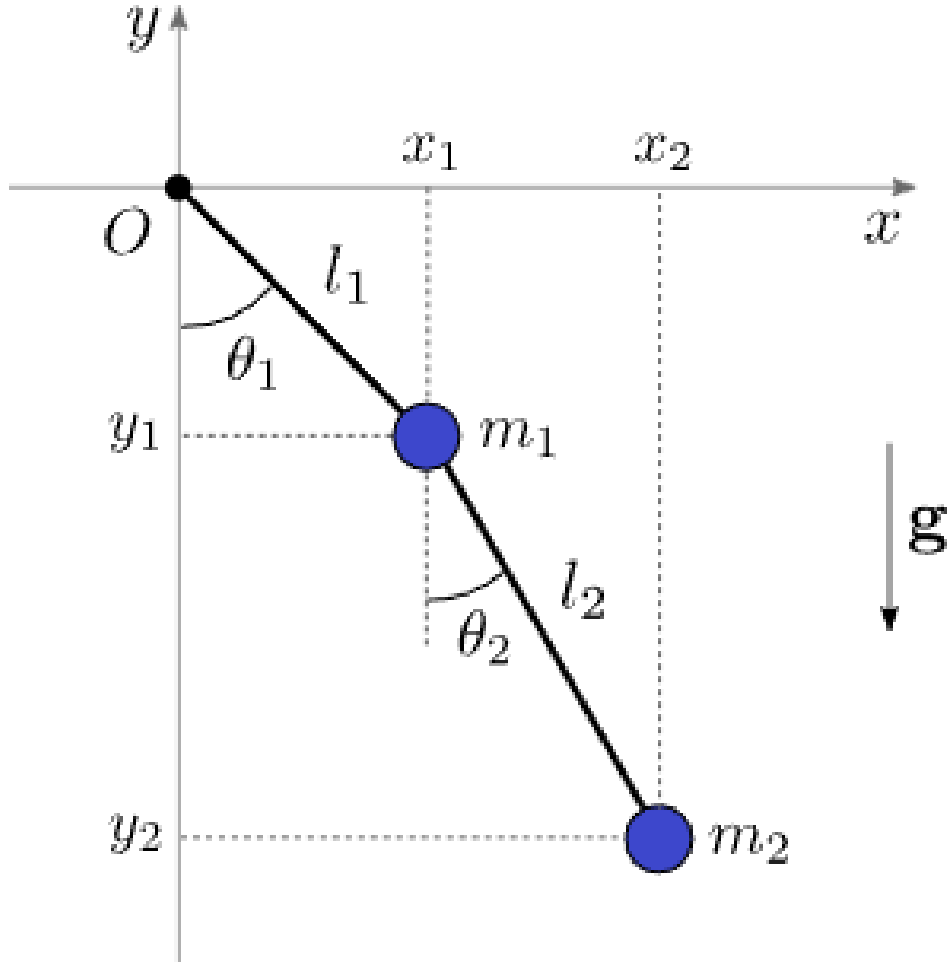
1 Introduction

This project studies the motion of a double pendulum using the Euler-Lagrange method to derive the equations of motion, followed by a fourth-order Runge-Kutta method (RK4) to solve the resulting system numerically.

2 Assumptions

- The masses are point masses.
- The rods are rigid and massless.
- The pendulum is ideal, with no loss from friction and no input from external forces.

3 Equations of Motion



3.1 Position

$$x_1 = l_1 \sin(\theta_1)$$

$$y_1 = -l_1 \cos(\theta_1)$$

$$x_2 = l_1 \sin(\theta_1) + l_2 \sin(\theta_2)$$

$$y_2 = -l_1 \cos(\theta_1) - l_2 \cos(\theta_2)$$

3.2 Velocity

$$\dot{x}_1 = l_1 \dot{\theta}_1 \cos(\theta_1)$$

$$\dot{y}_1 = l_1 \dot{\theta}_1 \sin(\theta_1)$$

$$\dot{x}_2 = l_1 \dot{\theta}_1 \cos(\theta_1) + l_2 \dot{\theta}_2 \cos(\theta_2)$$

$$\dot{y}_2 = l_1 \dot{\theta}_1 \sin(\theta_1) + l_2 \dot{\theta}_2 \sin(\theta_2)$$

3.3 Potential Energy

$$V = m_1 g y_1 + m_2 g y_2$$

Substituting y_1 and y_2 into the expression gives

$$V = m_1 g (-l_1 \cos(\theta_1)) + m_2 g (-l_1 \cos(\theta_1) - l_2 \cos(\theta_2))$$

$$V = -m_1 g l_1 \cos(\theta_1) - m_2 g l_1 \cos(\theta_1) - m_2 g l_2 \cos(\theta_2)$$

$$V = -(m_1 + m_2) g l_1 \cos(\theta_1) - m_2 g l_2 \cos(\theta_2)$$

3.4 Kinetic Energy

$$T = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2)$$

Substituting the velocities into the expression gives

$$T = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 (\cos^2(\theta_1) + \sin^2(\theta_1)) + \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2)$$

$$T = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 \left((l_1 \dot{\theta}_1 \cos(\theta_1) + l_2 \dot{\theta}_2 \cos(\theta_2))^2 + (l_1 \dot{\theta}_1 \sin(\theta_1) + l_2 \dot{\theta}_2 \sin(\theta_2))^2 \right)$$

$$T = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 \left(l_1^2 \dot{\theta}_1^2 (\cos^2(\theta_1) + \sin^2(\theta_1)) + l_2^2 \dot{\theta}_2^2 (\cos^2(\theta_2) + \sin^2(\theta_2)) + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 (\cos(\theta_1) \cos(\theta_2) + \sin(\theta_1) \sin(\theta_2)) \right)$$

$$T = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 (l_1^2 \dot{\theta}_1^2 + l_2^2 \dot{\theta}_2^2 + 2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2))$$

3.5 Lagrangian

$$L = T - V$$

Substituting T and V into the expression gives

$$L = \frac{1}{2}m_1l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\left(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right) - (-(m_1 + m_2)gl_1\cos(\theta_1) - m_2gl_2\cos(\theta_2))$$

$$L = \frac{1}{2}m_1l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\left(l_1^2\dot{\theta}_1^2 + l_2^2\dot{\theta}_2^2 + 2l_1l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right) + (m_1 + m_2)gl_1\cos(\theta_1) + m_2gl_2\cos(\theta_2)$$

3.6 Euler-Lagrange Equation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

3.6.1 Solving for the First Generalized Coordinate

$$q_1 = \theta_1$$

$$\frac{\partial L}{\partial \dot{\theta}_1} = (m_1 + m_2)l_1^2\dot{\theta}_1 + m_2l_1l_2\dot{\theta}_2\cos(\theta_1 - \theta_2)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) = (m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2l_1l_2\left(\ddot{\theta}_2\cos(\theta_1 - \theta_2) - \dot{\theta}_2(\dot{\theta}_1 - \dot{\theta}_2)\sin(\theta_1 - \theta_2)\right)$$

$$\frac{\partial L}{\partial \theta_1} = -m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - (m_1 + m_2)gl_1\sin(\theta_1)$$

$$(m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2l_1l_2\left(\ddot{\theta}_2\cos(\theta_1 - \theta_2) - \dot{\theta}_2(\dot{\theta}_1 - \dot{\theta}_2)\sin(\theta_1 - \theta_2)\right) - \left(-m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - (m_1 + m_2)gl_1\sin(\theta_1)\right) = 0$$

$$\ddot{\theta}_1 = \frac{-m_2l_1l_2\ddot{\theta}_2\cos(\theta_1 - \theta_2) - m_2l_1l_2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) - (m_1 + m_2)gl_1\sin(\theta_1)}{(m_1 + m_2)l_1^2}$$

3.6.2 Solving for the Second Generalized Coordinate

$$q_2 = \theta_2$$

$$\frac{\partial L}{\partial \dot{\theta}_2} = m_2l_2^2\dot{\theta}_2 + m_2l_1l_2\dot{\theta}_1\cos(\theta_1 - \theta_2)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) = m_2l_2^2\ddot{\theta}_2 + m_2l_1l_2\left(\ddot{\theta}_1\cos(\theta_1 - \theta_2) - \dot{\theta}_1(\dot{\theta}_1 - \dot{\theta}_2)\sin(\theta_1 - \theta_2)\right)$$

$$\frac{\partial L}{\partial \theta_2} = m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - m_2gl_2\sin(\theta_2)$$

$$m_2l_2^2\ddot{\theta}_2 + m_2l_1l_2\left(\ddot{\theta}_1\cos(\theta_1 - \theta_2) - \dot{\theta}_1(\dot{\theta}_1 - \dot{\theta}_2)\sin(\theta_1 - \theta_2)\right) - \left(m_2l_1l_2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - m_2gl_2\sin(\theta_2)\right) = 0$$

$$\ddot{\theta}_2 = \frac{-l_1\ddot{\theta}_1\cos(\theta_1 - \theta_2) + l_1\dot{\theta}_1^2\sin(\theta_1 - \theta_2) - g\sin(\theta_2)}{l_2}$$

3.7 RK4 Setup

The result is a system of two coupled differential equations, since the acceleration of one angle depends on the acceleration of the other angle. Before using RK4, the Python script rewrites the equations so that the accelerations are explicit.

The expression for $\ddot{\theta}_2$ is substituted into the expression for $\ddot{\theta}_1$, which gives an equation for $\ddot{\theta}_1$ depending only on the current angles and angular velocities. After $\ddot{\theta}_1$ is known, it is inserted back into the second equation to find $\ddot{\theta}_2$.

This gives the correct setup for RK4: at each RK4 step, and at each intermediate stage, both accelerations are evaluated from the current state before advancing the angles and angular velocities together.

3.8 RK4 Method

The numerical state is written as

$$\mathbf{y} = [\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2].$$

The derivative of this state is

$$\frac{d\mathbf{y}}{dt} = [\dot{\theta}_1, \ddot{\theta}_1, \dot{\theta}_2, \ddot{\theta}_2].$$

The RK4 method advances this state forward in time by evaluating the derivative four times during each time step and combining the results.

3.9 Energy Check

Since the pendulum is assumed to be ideal, the total mechanical energy should be constant. The kinetic energy, potential energy, and their sum are therefore tracked during the simulation. Any change in the total energy is numerical drift from the RK4 approximation.

4 Conclusion

The Euler-Lagrange method was used to derive the coupled equations of motion for the double pendulum. These equations were then rewritten into a form suitable for numerical simulation and solved using RK4. Energy was also tracked to check how well the numerical solution conserved the ideal system energy.